Name Kamalesh V.S

Standard _____ Division _____ Roll 1114

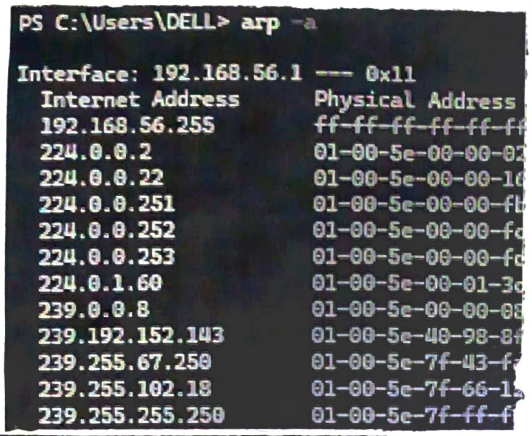
Subject _____

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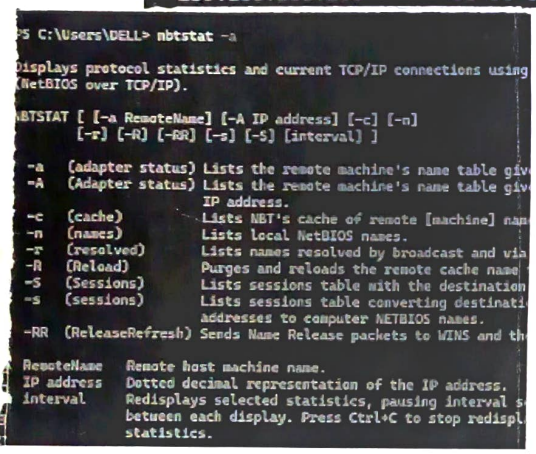
NOTES

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```
PS C:\Users\DELL> arp -a

Interface: 192.168.56.1 --- 0x11
  Internet Address      Physical Address
  192.168.56.255        ff-ff-ff-ff-ff-ff
  224.0.0.2             01-00-5e-00-00-02
  224.0.0.22            01-00-5e-00-00-16
  224.0.0.251           01-00-5e-00-00-fb
  224.0.0.252           01-00-5e-00-00-fc
  224.0.0.253           01-00-5e-00-00-fd
  224.0.1.60            01-00-5e-00-01-3c
  239.0.0.8             01-00-5e-00-00-08
  239.192.152.143       01-00-5e-40-98-8f
  239.255.67.250        01-00-5e-7f-43-fa
  239.255.102.18        01-00-5e-7f-66-12
  239.255.255.250       01-00-5e-7f-ff-fa
```



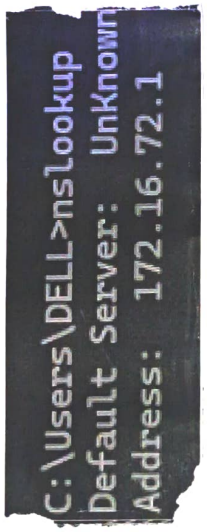
```
PS C:\Users\DELL> nbtstat -a

Displays protocol statistics and current TCP/IP connections using
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table giv
  -A   (Adapter status) Lists the remote machine's name table giv
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] nam
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via
  -R   (Reload)         Purges and reloads the remote cache name
  -S   (Sessions)       Lists sessions table with the destination
  -s   (sessions)       Lists sessions table converting destinati
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and th

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval s
               between each display. Press Ctrl+C to stop redispl
               statistics.
```



```
C:\Users\DELL>nslookup
Default Server:  UnKnown
Address:  172.16.72.1
```

Exp. No : 1

16/07/2026

EXP. NO. 1

BASIC NETWORKING COMMAND

AIM :

To implement Basic Networking Commands in linux and windows operating system.

BASIC NETWORKING COMMANDS :

- arp -a :

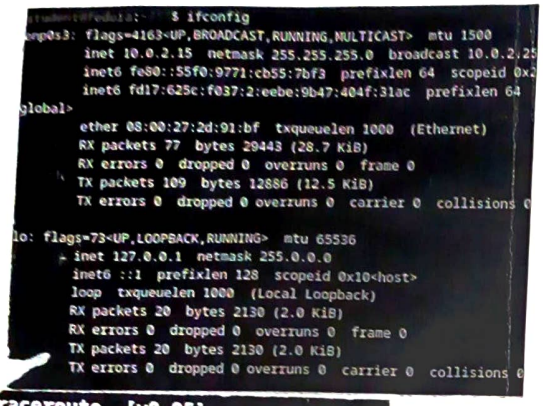
ARP is short form of address resolution protocol.

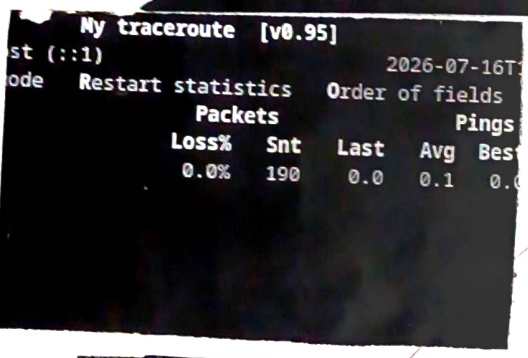
- hostname :

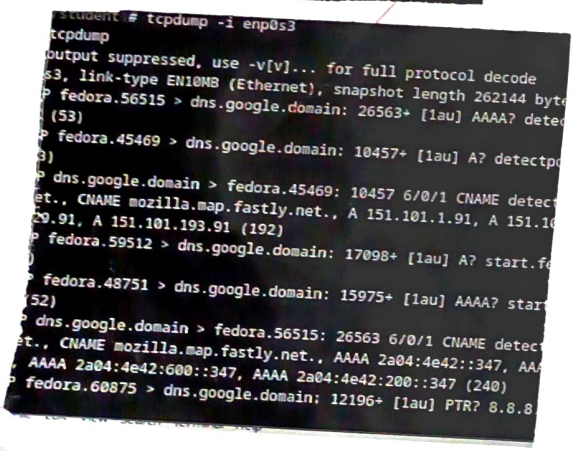
This is the simplest of all TCP/IP commands. It simply displays the commands.

- ipconfig /all :

This command displays detailed configuration, information about the connection including Router, Gateway, DNS, DHCP.







- netstat - a :

This command helps solve problem with NetBIOS name resolution for TCP/IP connection. It is a command line tool for monitoring network incoming and outgoing.

- nslookup :

name server lookup is a tool used to perform DNS lookups in display DNS details, such as IP address of a particular computer.

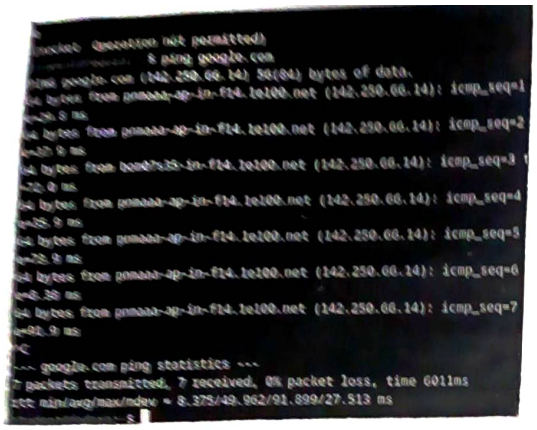
eg : nslookup www.google.com

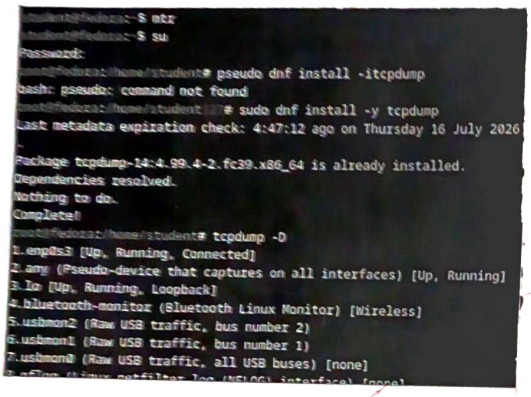
- Pathping :

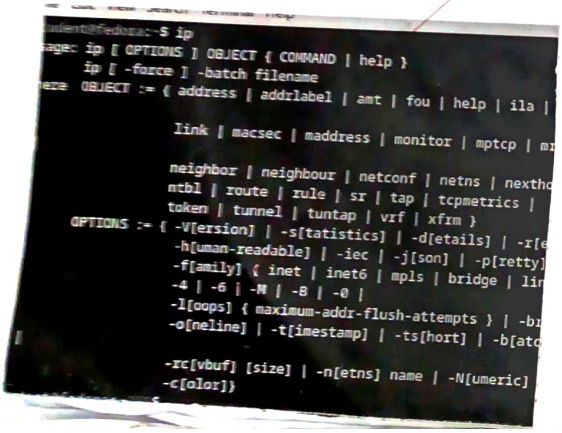
Pathping is unique to window's and is basically a combination of tracert commands.

- ping :

ping is a command to test connectivity needs.







- Route :

Route Commans is used to show / manipulate the IP routing.

LINUX NETWORKING COMMANDS :

- ip :

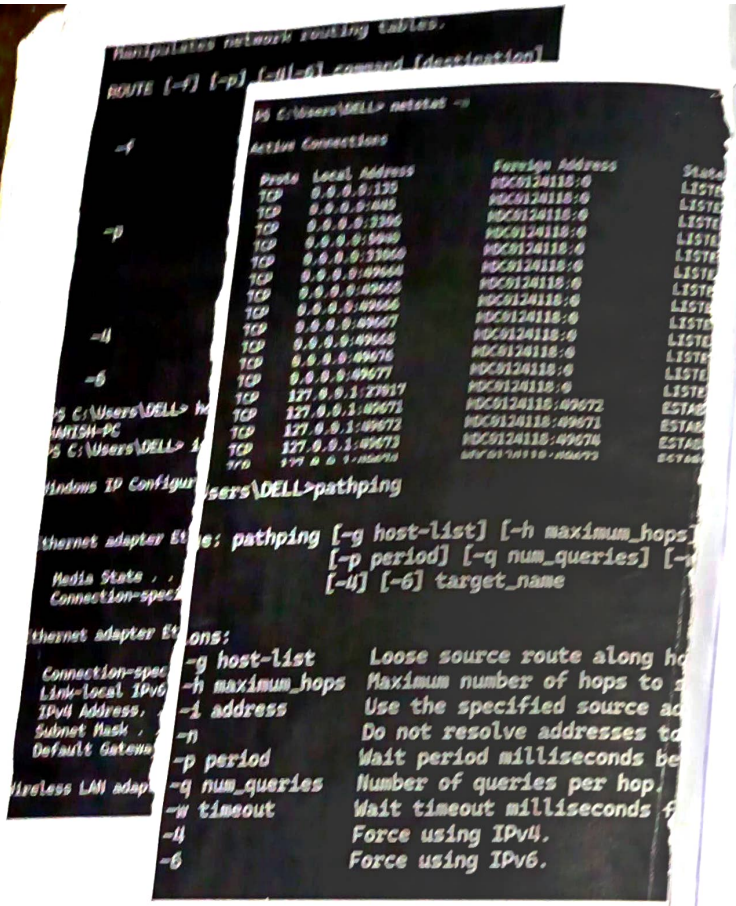
The ip command can show address information, manipulate routing, plus, display network.

ip< options ><object><command>

- To show the Ip addresses assigned to an interface on your server.
- To assign an Ip to an interface like enps03.

2) Ifconfig

The ifconfig command was a staple in many sysadmin's tool belt for configuring and trouble - shooting network.



- mtr

MTR is a program with a command-line interface that serves as a network diagnostic and troubleshooting tool. This command combines the functionality of the ping and traceroute commands.

- tcpdump

The tcpdump command is designed for capturing and displaying packets.

- ping :

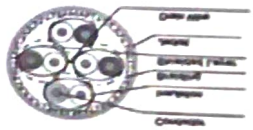
Ping is a tool that verifies IP-level connectivity to another ICP/IP Computer by sending Internet control Message protocol.

Result:

Basic networking Commands in both linux and windows operating System is successfully executed & verified









Exp-2

STUDY OF DIFFERENT TYPES OF NETWORK CABLES

Aim :

To study and understand different types of network cable, To make your own ethernet cross over cable / Straight cable.

Different Types of Network Cable:

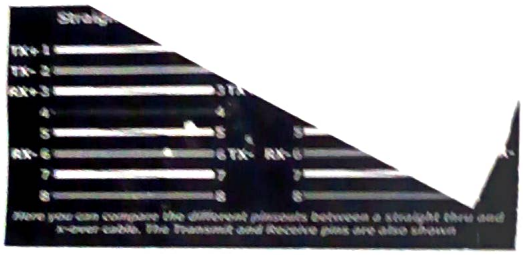
1) Unshielded Twisted Pair Cable
2) Shielded Twisted pair Cable
3) Coaxial Cable
4) Fibre optic Cable.

UTP :

- In Unshield Twisted pair Cable has three categories
- Category 3 has 10bps data Transmission Capacity
- Category 5 has upto 100 mbps data Transmission
- Category 5e has 1Gbps has Capacity.







Here you can compare the different pinouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

Cable type	Category	Maximum length (Ethernet)	Advantages / Disadvantages	Application
UTP	Category 3	10 Mbps	Advantages • Cheaper in cost • Easy to install	10 Base - T ethernet
	Category 5	Up to 100 Mbps	- they have small overall diameter	Fast ethernet, Gigabit
	Category 5e	1 Gbps	More prone to EMI	
STP	Category 6, 6a	10 Gbps	Advantages • Shielded • Faster • Less prone to noise	• Gigabit ethernet • Widely used in datacenter
Coaxial Cable	RG-6 RG-59 RG-11	10 - 100 Mbps	Advantages • High bandwidth • Immune to interface	• Speed of signal is 500 m • High Speed
Fibre optics Cable	Single mode Multi mode	100 Gbps	Advantages • High Speed • High bandwidth • High Security Disadvantage • Expensive	• Maximum distance of fibre optics Cable is around 100 m

Make your own Ethernet Cross-Over

Tools and parts needed:

- Either Cabling Cat 5e is certified for gigabit, support but Cat 5e Cabling work as well
- A Crimping tool
- Two RJ45 plugs
- Optional two plug shield

Step 1:

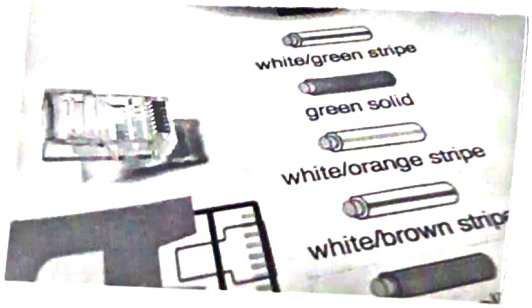
To start construction of the device begin by threading

Step 2:

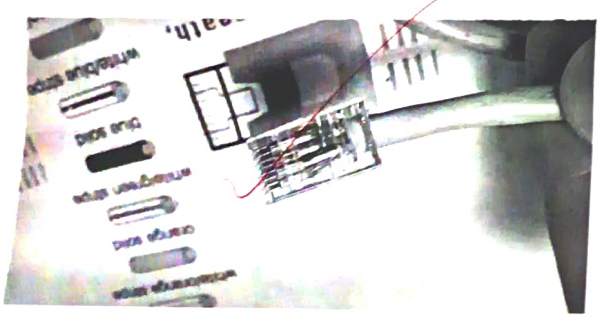
Next strip approximately 1.5 cm of cable shielding from both ends.

Step 3:

After you will need to untangle the wires, there should be four twisted pairs.







Step 4:

Once the order is correct. Bunch them to gather in a line and if there are any that stick out farther than other strip them back to create an even level.

Step 5:

Next push the cable right in. The not chatthendo to be just over the cable shielding and if it isn't that means that you stripped off too much shielding.

Step 6:

After wires are securely sitting inside the plug insert it into crimp - ingo to go and push

Step 7:

Lastly repeat for theother endusing diagram to test it plug it in attempt to connect two devices directly.

Result :

Thus, different types of network cables were studied successfully and straigh through and cross over ethernet cable were tested successfully.

EXP 3
23/7/26

UNDERSTANDING THE CISCO PACKET TRACKER ENVIRONMENT

Aim:

To understand the cisco packet envisonment and design a simple Computer network.

Steps:

* Open the cisco packet Tracker
* Obseeve the UI including the Menu bar, Main toolbar, Workspace, Common tools bar, Network corponent.
* From the network Component box, Select the required network device such as PCs, Hub.
* Drag and Drop space in workspace
* verify all links are active
* Save the netwo rk topology.

Result:

The Cisco packet Tracer environment was successfully explored, and a simple network was designed. The interface Components, device connections, IP configuration were successfully understood using both Realtime & Simulation Mode.



Exp: 3b1
23/07/26

HUB

Aim:

To study the behavior of a hub observing it forward data packet using cisco packet Tracer.

Steps:

* Open Cisco Packet Tracer
* Drag and Place 1 Hub and 4 Generic PCs into the workspace
* Connect all PCs to hub using Copper Straight-Through Cables
* Select Add simple PDU from common tools bar
* Send a packet from one PC to another PC
* Switch to simulation Mode and observe how fast hub forward the packet.

Result:

The hub broadcast the received packet to all connected devices. Every Pc receives the pack only the destination pc accept it.

EXP 3b2

SWITCH

To study the behavior of a switch by observing how it forward data packet using cisco packet.



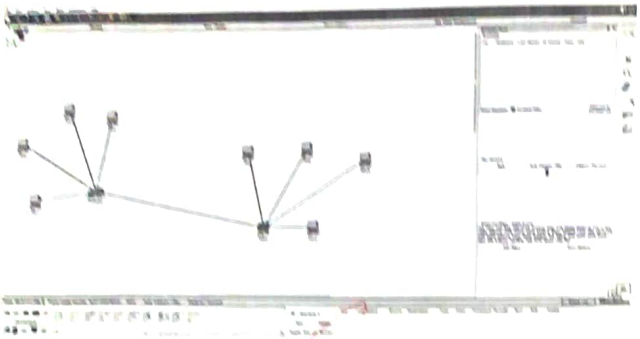




* Open cisco packet Tracer
* Drag and place 1 switch and 4 Generic PCs into workspace
* Connect all PCs to all switch using copper cable
* Select Add simple PDU from commol - tool bar
* Send a packet from one PC to another
* Switch to simulation mode and observe

Result:

The switch forward the packet only to the destination PC by using its MAC address table. This reduces unencessary network traffic.



EXP: 8&9

HUB AND SWITCH

Aim:

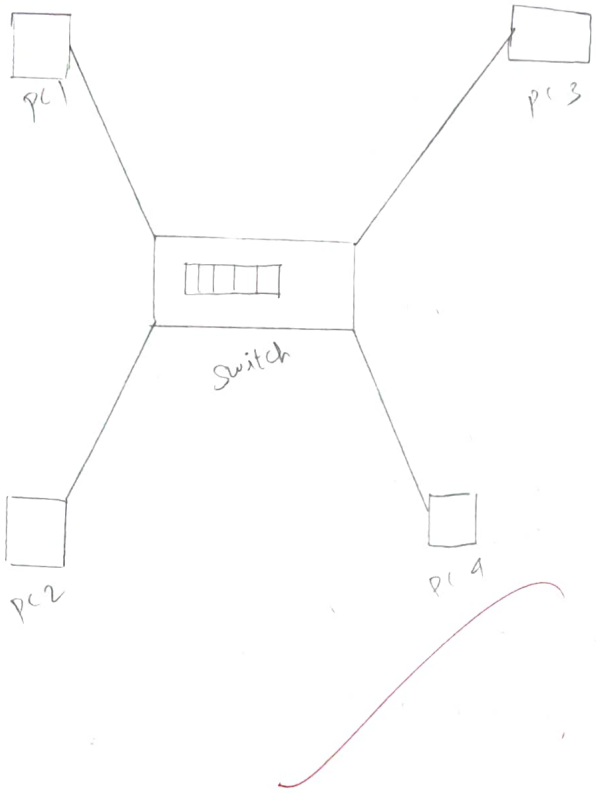
To compare the packet forwarding behavior of a hub and switch using cisco packet Tracer.

Steps:

* Create one network using a hub and another using switch.
* Configure IP address for all PCs in both network.
* Send a simple PDU between two pcs in each network.
* observe packet transmission in simulation mode.
* Compare how the hub and switch forward packet.

Result:

The hub broadcast packet to all connected devices, whereas the switch forwards only to packet. Therefore Switch provide better performance.



EXP 4
30/7/23

CONFIGURATION OF LAN USING SWITCH AND ETHERNET

Aim:

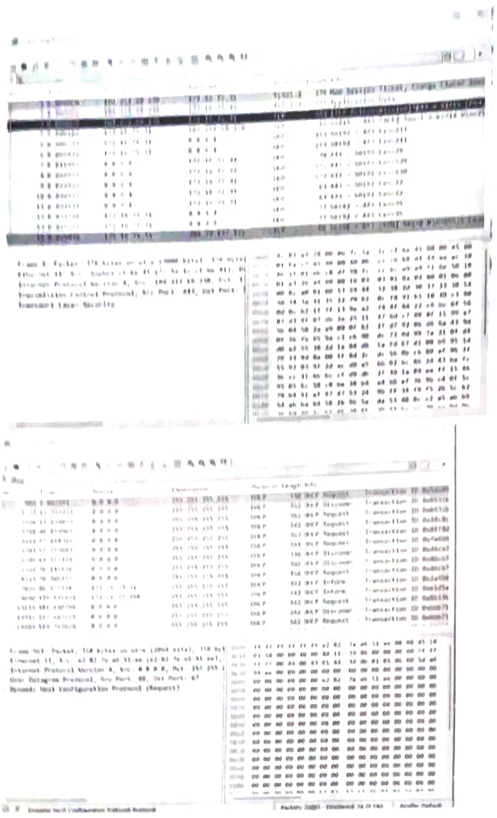
Setup and configure a LAN using a switch and ethernet cable

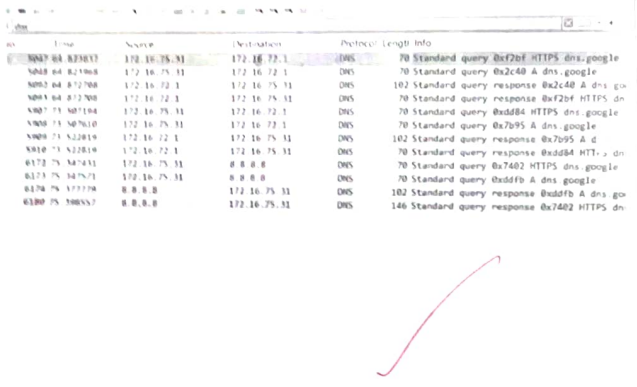
Steps:

1. Plan and design an appropriate network topology talking into account network.
2. you can take 4 computer, a switch with 8, 16 or 24 ports.
3. connect your computer to network Switch via ethernet cable.
4. Assign IP address
5. configure switch network.
6. chek the connectivity
7. Try access the shared folder.

Result:

Stepup and configuration of a lan using switch and ethernet cables is successful.





EXP: 5 EXPERIMENT ON PACKET CAPTURE TOOL WIRESHARK:

AIM:

To understand the feature of wireshark as a packet capture tool and understand encapsulation of information.

PACKET SNIFFER:

* Capture packet sent and received by computer
* Display protocol information of captured packet.

SNIFFER TOOLS:

* TCP dump - Command line packet
* Wireshark - Graphical packet analyz

USES OF NETWORK:

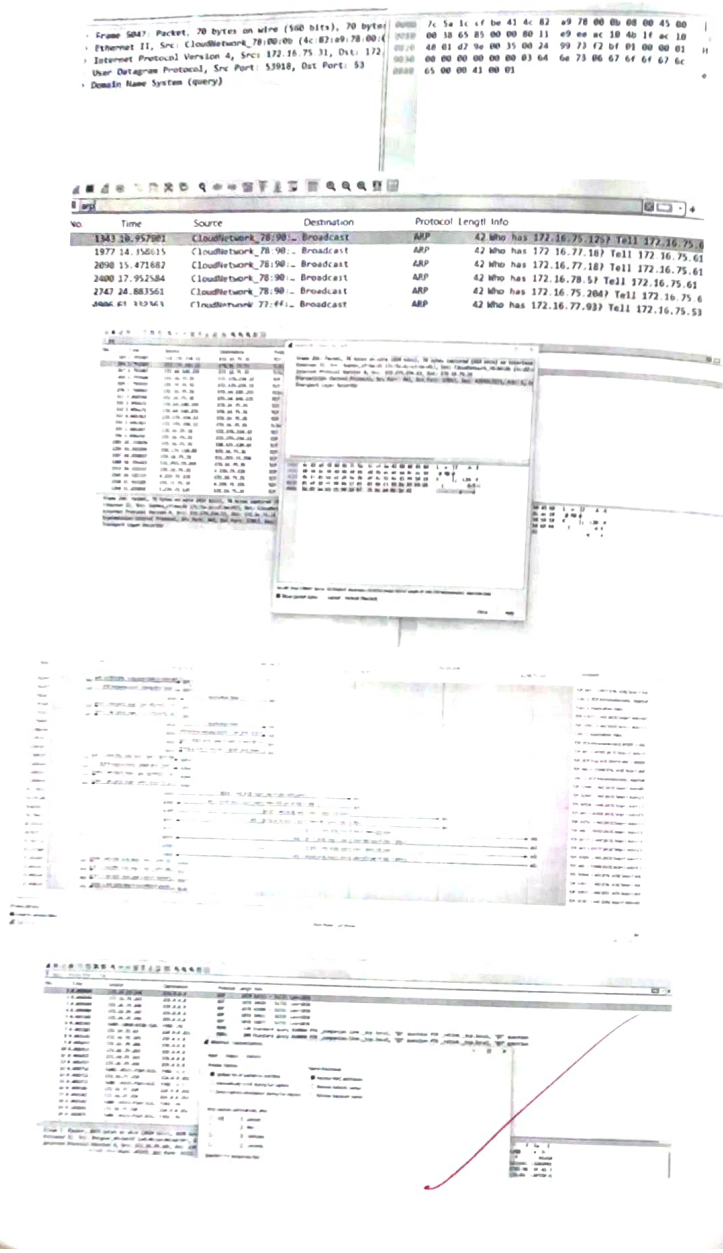
Capture network Traffic
Analyzes protocol.

FLOW GRAPH:

Displays the communication flow between devices making packing.

CAPTURE 1000 PACKETS:

* Open the Capture file
* Apply the TCP || UDP filter
* Inspect and view the flow graph
* Apply the DNS filter



* Inspect and view flow graph and save the capture.

RESULT:-

Thus the feature of wireshark and implement of different filter was Succesfully completed.

EXP 6
23/7

ERROR DETECTION AND CORRECTION USING HAMMER CODE

AIM:

To write a program to implement error detection and correction using hamming code.

ALGROTHM:

* Start the program
* Read the number of 4-bit data blocks (n) forming input
* For each 4-bit data block D = d1, d2, d3, d4 perform Hammer encoding
* Compute the parity bits at position

P1 = d1 XOR d2 XOR d4

P2 = d1 XOR d3 XOR d4

P4 = d2 XOR d3 XOR d4

* At the receiver, recompute the check bits from received code:

C1 = P1 XOR d1 XOR d2 XOR d4

C2 = P2 XOR d1 XOR d3 XOR d4

C4 = P4 XOR d2 XOR d3 XOR d4

* Compute error position
* Extract the original 4 bits
* Repeat step 3 to 8 for all blocks
* Stop the program

PROGRAM:

```
def encode(d):
    h = [0] * 7
    h[2] = d[0]
    h[4] = d[1]
    h[5] = d[2]
    h[6] = d[3]
```

OUTPUT :

```
==HAMMING CODE - ERROR DETECTION
            AND CORRECTION==

Enter number of 4 bit data block
in stream: 2
--- Block 1 ---
Enter 4 data bit (space seperated): 1 0 1 1
Original Data bits: 1011
Encoded Hamming Code: 0110011
Received Code: 0100011
Checking for errors....
   Error detected at bit position 3
   Correcting.....
Corrected Code: 0110011
Recovered data bits: 1011
--- Block 2 ---
Enter 4 data bits: 0110
Original Data bits: 1100110
Encoded Hamming Code: 11000110
Enter bit position: 6
Received Code: 1100100
```

```
def detect_and_correct(h):
    c1 = h[0] ^ h[2] ^ h[4] ^ h[6]
    c2 = h[1] ^ h[2] ^ h[5] ^ h[6]
    c4 = h[3] ^ h[4] ^ h[5] ^ h[6]
    error_pos = c1*1 + c2*2 + c4*4
    if error_pos != 0:
        printf(f" Error detected at bit:
{error_pos} -> correcting... ")
    else:
        print (" No error detected")
def main():
    print ("== HAMMING CODE - ERROR
          DETECTION AND CORRECTION ==")
    n = int(input("Enter number of 4-bit
                  data block: "))
    h = encode(d)
    printf(" Original data Bits : {to_str(d)}")
    printf(" Encoded Hamming Code (7 bits )")
    for b in range(1, n+1):
        ~~Printf "Original Data bit~~
        d = list(map(int, input("
Enter 4 data bits (space seperated):
).split()))
```

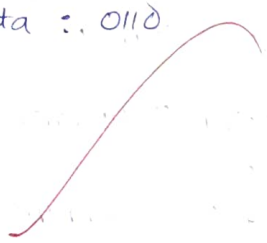
Checking for errors....

Error detected at bit position: 6

Correcting.....

Corrected code : 1100110

Recovered data : 0110



```
if 1 <= pos <= 7:
    h[pos - 1] ^= 1
    print(f"Transmitted code (with error)
else:
    print("Transmitted code (without error)

print("Received code : {to_str(h)}")
print("Checking for error....")
detect_and_correct(h)
print(f" corrected code: {to_str(h)}")
if __name__ == "__main__":
    main()
```

Result:

The program to implement error detection and correction using the hamming code concept was successfully written in python, executed and verified using a sample input data stream.

OUTPUT:

= SLIDING WINDOW PROTOCOL - FLOW CONTROL

Enter total number of frames: 8
Enter Sender's window size: 4
Enter frame number to simulate
lost (
(0 if none - to no loss): 5

Sender window: [Frame 0 - Frame 3]

Node A ⟶ Sending frame 0
Node A ⟶ Sending frame 1
Node A ⟶ Sending frame 2
Node A ⟶ Sending frame 3
Node B ⟶ Received frame 0,
Sending ACK 0
Node B ⟶ Received Frame 1,
Sending ACK 1
Node B ⟶ Received frame 2,
Sending ACK 2

FLOW CONTROL AT DATA LINK LAYER USING SLIDING WINDOW PROTOCOL

Aim:

To write a program to implement flow control data link layer using the Sliding window.

Algorithm:

1) Start the program
2) Read the total number of frame to be sent (N), the sender's window size
3) Initialize base = 0 and nextSeq = 0
4) Repeat while base < N
5) Repeat Step 4 unit all N frames have been transmitted and acknowledged
6) Stop the program.

Program:

```
def Simulate (total_frame, lost_frame
    base = 0
    next_seq = 0
    while base < total_frames
    window_end = min(base + window
                 - size - 1)

    print ("\n.....")
    print (f" Sender Window: [ frame ")
```

Sender window: [Frame 4 ~ Frame 7]

Node A → Sending frame 5
Node A → Sending frame 6
Node A → Sending frame 7
Node B → Received frame 4, Sending ACK 4
Node B → Received frame 5, Sending ACK 5
Node A → ACK 7 received. Sliding window forward

All 8 frames successfully transmitted from Node A to Node B.

```
print(f" {frame [next_seq] LOST IN}")
next_seq += 1
If lost_frame is not None and box
    <= lost_frame
print(f" Node B --> Frame {lost_frame}
print(f" Node A --> Time Out! Resending
next_seq = lost_frame
lost_frame = None
 Continue
for f in range (base, next_seq):
    print(f"Node B --> Received
                        Frame {f}
Sending ACK {f}")
    print(f" Node A }
    If __name__ == "__main__"
        main()
```

Result:

The program to implement flow control at the data link layer using the Sliding window protocol was successfully written in python executed and verified by simulating the flow of frames.

Output :

ARP :

Sud : nmap -AR -sn

Starting Nmap

Nmap Scan report from 10.10.210.1

Host is up

Nmap done

Icmp :

Amap -ps -Sn 10.10.210.0/24

Starting Nmap

Nmap Scan report for 10.10.210.6

Host is up

Nmap. done

TCP :

nmap -ps 230 -Sn 10.10.210.0/24

Sudo nmap PA 80.Sn. 10.10.210/24

EXP.8
3/7

NMAP TO DISCOVER LIVE HOST USING NMAP SCANS

Aim :

To study the working of the Nmap tool and discover live host in a network using different host discovery techniques such as ARP Scanning.

Steps :

1) Open the TryHackMe platform and select the Nmap room.

2) Study the network topology and identify the Source and destination hosts.

3) Use ARP request to discover the MAC address.

Host Discovery Techniques :

i) ARP Scan :

* Uses ARP requests to devices on the same Subnet.

* Hosts responding with ARP replies are considered active

ii) Host TCP / UDP Scan :

* TCP SYN ping (Sends SYN packets to selected ports

* UDP ping checks host availability through ICMP port unreachable responses

observations :

ARP Discovery :

* Active hosts respond with ARP repli
* Number of devices discovered using ARP

ICMP Discovery :

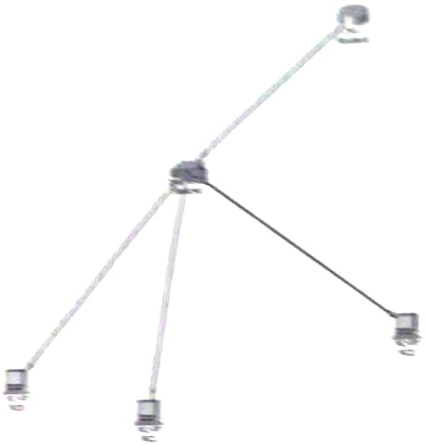
* ping requests help identify active hosts.
* ICMP scans may fail if firewall block ICMP traffic.

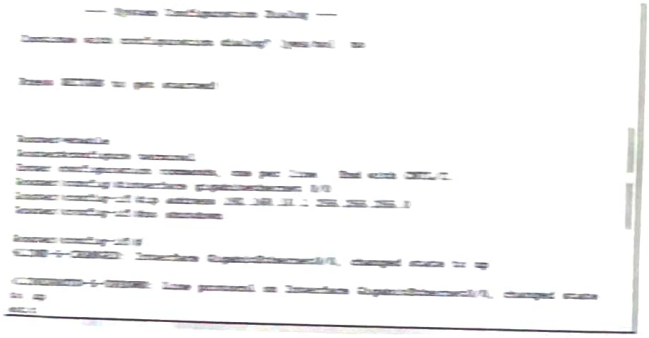
TCP/UDP Discovery :

* TCP SYN ping doesnot require a privileged account.
* TCP ACK ping requires a privileged account.

Result :

Thus live hosts in the network were successfully using different Nmap scanning techniques. ARP scanning was found to be effective for finding, while ICMP and UDP scanning techniques.





n.9a
2081

Implementation of Subnetting in CISCO PACKET TRACER

AIM:

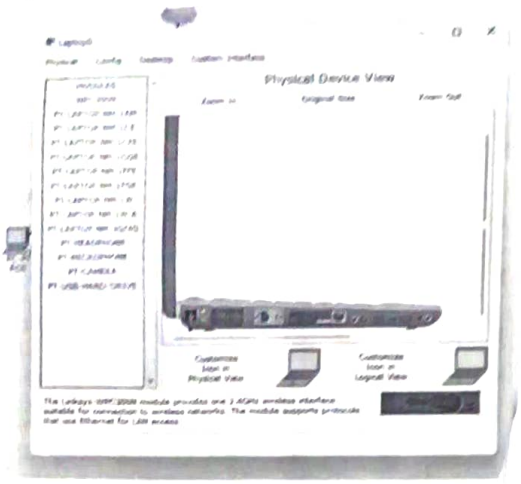
To implement subnetting by dividing a large network into multiple smaller subnetwork according to required number of host.

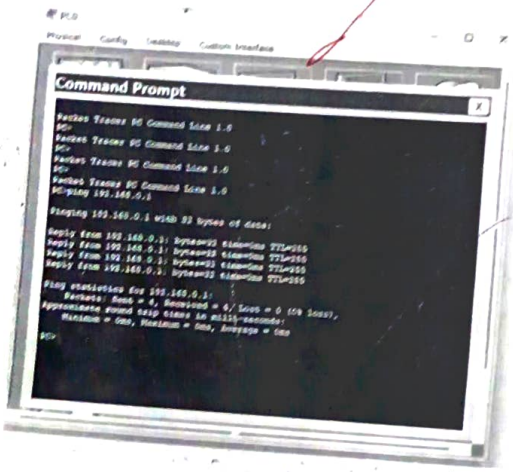
Steps:

1) Open cisco packet Tracer and create a new project
2) Design the network topology by placing required network device
3) Select a suitable network address
4) Calculate number of Subnet.
5) Determine no. of. hosts needed
6) Calculate Subnet mask
7) Divide the network into multiple
8) Assign a separate network address
9) Verify the subnet calculation
10) Save the network topology.

Result:

Thus multiple subnets were successfully created according to the required number of hosts, and the network address was efficiently divided.





Exp 9b
7/9

ASSIGN STATIC IP ADDRESS ALL SUBNET AND CONNECT

Aim :

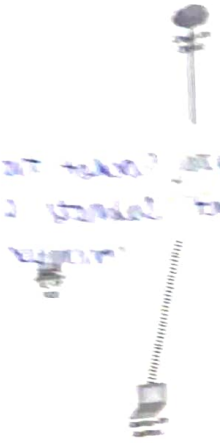
To assign Static IP addresses to all device in different subnet.

Steps :

1) Place a router in network topology
2) Connect the router to switch
3) Configure the router interfaces
4) Assign an IP address and subnet.
5) Configure the Subnet mask for each.
6) Assign the appropriate default gateway.
7) Configure the Subnet mask for each.
8) Assign the appropriate gateway.
9) Verify the IP configuration of all devices
10) Save the router configuration.
11) Check the network topology.

Result :

Thus static IP address were successfully assigned to all devices and communication between different subnet was established.



Simulate Packet Transmission Across the Subnet

...transmission
...when a
...is used for ...ication.

Steps:

1) Open Simulation Mode in Cisco Packet Tracer.
2) Select a source device from one Subnet.
3) Select a destination device from another Subnet.
4) Generate a packet using PDU
5) Observe the movement of the packet through network.
6) Analyze how router ...ceives ... forwarding process
... the packet transmission
... command ... connectivity
... successful delivery of packet.

Result:

Thus packets were successfully transmitted between different subnets through the router.

Exp 9c
Ex No: 11

When Subnets are Not Connected through a Router

Aim:

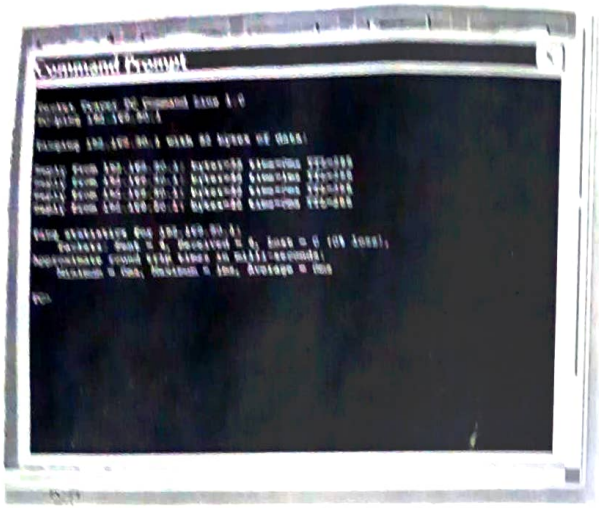
To analyze packet transmission between different subnets when no router is available.

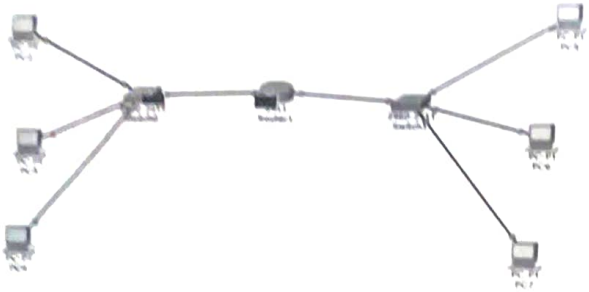
Steps:

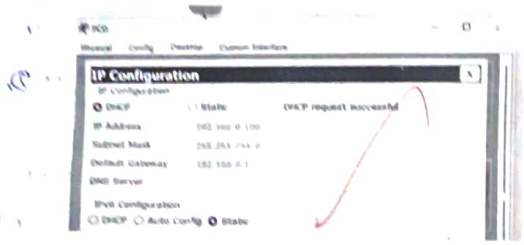
1) Remove the router from topology
2) Open Simulation Mode
3) Select devices belonging to different subnets.
4) Generate packets using the simple PDU tool.
5) Observe the packet transmission
6) Perform a ping test.
7) Analyze the packet flow
8) Observe the simulation events.
9) Identify the point at which packet transmission fails.
10) Compare the result with the network that uses a router.
11) Record the observation

Result:

Thus communication between devices located in different subnets failed when a router was not used.







Exp 10 Internetworking with routers in cisco packet tracer simulator

Aim :

To design and configure a simple internetwork using a router.

Procedure :

1. Internetworking Using a router

1. Open cisco packet Tracer and add one router.
2. Connect both pcs to the router using copper straight.
3. Configure the router.
4. Configure pc0 and pc1
5. Test connectivity using simple PDU.

2. Wireless Router, DHCP Server and Internet cloud.

1. Add a wireless router, pc, laptop
2. Connect the device using appropriate Cables
3. Set the wireless router SSID to homenetwork and enable DHCP
4. Connect the laptop to HomeNetwork and configure the PC.
5. Configure the cisco.com

The interfaces was successfully configured in Cisco Packet Tracer. Ecuta Communication between connectivity, DHCP and wireless connectivity verified successfully





Aim :

To Simulate and configure RIP (Routing Information protocol) using Cisco Packet Tracer.

Procedure :

1) Open cisco Packet Tracer
2) Place three routers and two PCs.
3) Connect the three routers in a triangular topology
4) Connect PC0 to Router 0 and PC1 to Router 2.
5) Configure the Ip addressess of all router interface and PCs
6) Enable the interfaces using no Shutdown.
7) Configure RIP on each router.
8) Advertise the directly connected network using network Command
9) Verify the routing table
10) Test Connectivity between PC0 & PC1

Result :-

Thus RIP was successfully Simulated Using Cisco packet Tracer.

TCP Echo server listening to port 5000
Client connected: /127.0.0.1:50303
Received: hi
Received: hi i am kamalesh

Type message to send (type 'bye' to quit)
Server: Echo hi
hi i am kamalesh
Server: Echo: i am kamalesh

Program:
Import j

Exp 12a
17/9

END TO END COMMUNICATION AT TRANSPORT LAYER

a) Echo Client - Server Using TCP

Aim:

To implement an echo client server communication program using TCP socket programming, where the server receives a message from client.

Procedure:

1) Create a TCP socket at server
2) Blind the server socket to IP
3) Make the server listen for incomming client
4) Accept the client connection
5) Create a TCP socket
6) Connect the client
7) Client sends message to server
8) Client receives a message
9) Close the socket after communication

Result:

Thus the TCP echo client-server communication was successfully implemented and message sent by client was received back.

UDP echo server listening to Port

Received from 127.0.0.1.6582 21 -> hi

Received from 127.0.0.1.6582 21 -> hi how are you

Type message to send

hi

Server: Echo: hi

hi how are you

Server: Echo: hi how are you

UDP socket is

Aim:

To implement client server communication using UDP sockets.

Procedure:

* Create a UDP socket at server
* Bind the server socket to an IP address and port number.
* Create a UDP socket at client
* Server receives message using recvfrom
* Server processes the received message
* Server sends a response
* Display the received message.
* Close the socket after communication

Result:

Thus UDP socket communication using socket programming was successfully implemented.

Enter your Name:
Kamalesh

Welcome, Kamalesh !

Sa

TCP	0.0.0.0.5000	0.0.0.0	Listening
TCP	127.0.0.1.5000	127.0.0.1	Established
TCP	127.0.0.1.50503	127.0.0.1	Established
TCP	[::]:5000	[::]:0	listening

EXP IV
17/9

CHAT PROGRAM USING SOCKET PROGRAMMING

Aim :

To implement a chat application using socket programming that allows a client and server.

Procedure :

* Create a socket at the server
* Bind the socket to an IP address
* Make server listen for connections.
* Accept client connection.
* Create a socket at client.
* connect the client
* Server enters a reply
* Client receives and displays the reply.
* Repeat the send and receive process
* Terminate & close the sockets.

Result :

Thus a client-server chat program using socket programming was successfully implemented.

Program:

```
import java.net.InetAddress;
import java.util.Scanner;
public class myping {
    public static void main(String args[]) {
    Scanner sc = new Scanner(System.in);
    String host = sc.nextLine();
    my {
    InetAddress = InetAddress.getByName
    System.out.print("pinging" + host);
    long start time = System.currentTime();
    If (reachable) {
      System.out.println("Reply Out");
    else {
    System.out.print("Reply from" + getHo
    System.out.println("Host is not reached
    }
    } catch (Exception e) {
        System.out.println("Error");
    }
    sc.close();
} }
```

IMPLEMENT YOUR OWN PING PROGRAM

Aim:

To implement a ping program using socket programming to check whether a destination is reachable.

Procedure:

* Create a socket for communication with the destination host.
* Get the IP address of the destination host.
* Construct an ICMP Echo Request.
* Add the required ICMP header and sequence number.
* Record the time before sending
* Wait for the reply.
* Calculate the Round-Trip time
* Repeat the process for the required number of packet.
* Display the final statistics such as packet sent, received, lost.

Result:

Thus the Own Ping Program was successfully implemented to check the host reliability.

Source IP: 172.16.72.225

Destination IP: 239.0.0.5

Protocol: 17

Packet length: 1038

Implement packet sniffing using raw sockets

Aim:

To implement packet sniffing using raw sockets in Python and capture network packets.

Procedure

1) Open VS code and create a python file name packet_sniffer.py
2) Import the required socket and struct libraries.
3) Create and configure a raw socket using local IP address.
4) Run VS code as Administrator and execute the program.
5) Generate network traffic using ping www.bbc.com
6) Capture and display the source IP, destination IP.

Result:

Thus, Packet sniffing was successfully using raw sockets in Python, and network packets were captured and their details were displayed.